

Dayton Arrow University 2026 Seminars

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Session 1

Diodes Incorporated

Solve common design problems with programmable, mixed-signal uASICs from Diodes Inc

Learn how to save board space, power, and costs with Diodes' configurable uASICs solutions for industrial and consumer applications. In this session we will address common design needs like power sequencing, watchdog supervisors, waveform generators, voltage monitors, OVP/UVF and OCP functions, and the implementation of simple and complex logic functions in a programmable, mixed-signal ASIC. We will show design tool examples and demonstrate how simple it is to implement a solution using uASICs from Diodes Inc.

Intel

Edge Intelligence Redefined: Maximizing AI Inference Efficiency with Intel® Core™ Ultra

Description: As artificial intelligence transitions rapidly from cloud data centers to the industrial and embedded edge, developers face a severe three-way constraint: processing massive AI workloads within rigid power, size, and cost budgets (SWaP-C). Traditional edge architectures often force an uncomfortable compromise—either relying on expensive, power-hungry discrete accelerators or suffering from high latency and poor throughput on standard CPUs. This technical session introduces the Intel® Core™ Ultra processor family and demonstrates how its breakthrough disaggregated, tile-based architecture redefines edge AI inferencing. Attendees will explore the synergy between the CPU, the integrated AI-centric GPU, and the dedicated NPU (Neural Processing Unit). We will dissect how this specialized hardware harmony intelligently distributes diverse AI workloads—such as computer vision, predictive maintenance, and local generative AI—to drastically reduce system latency and system power consumption. Finally, the session will highlight how consolidating these capabilities onto a single silicon footprint eliminates the cost and complexity of discrete hardware accelerators, enabling ultra-compact, fanless embedded designs.

Phoenix Contact

Overcoming Connectivity Challenges in Robotics

Robotics relies on robust connectivity for real-time control, data exchange, and system integration. This encompasses internal electrical and electronic connections as well as communication between robots and external devices. Performance is often constrained by latency, bandwidth limitations, and network instability—particularly in autonomous vehicles, industrial automation, and multi-robot environments. Advances in connector design and emerging technologies offer improvements, yet achieving reliable, low-latency connectivity in dynamic conditions remains a critical challenge for fully autonomous and collaborative robotics.

STMicroelectronics #1

Introduction to ST's VL53L9 3D Time-of-Flight LiDAR Sensor

Discover the capabilities of STMicroelectronics' new VL53L9 direct Time-of-Flight (dToF) 3D LiDAR sensor in this high-level technical overview. This introductory session explores the sensor's key architecture including its 2,268 resolution zones and high-speed MIPI/I3C interface options. Ideal for system architects and hardware engineers, this course provides the essential technical foundation needed to evaluate the VL53L9 for next-generation spatial sensing, drones, robotics, and user-detection applications.

Session 2

Lattice

Utilizing FPGA's as Companion chips in embedded applications

As MCU/MPU demands grow in embedded and edge AI applications, designers can offload tasks to small, low-power, low-latency FPGAs. This seminar shows how adding FPGAs expands system headroom and enables new features without requiring major changes to existing processors from Qualcomm, NXP, ST, Nvidia, and others.

NXP

From Sensor to Action: Designing Reliable Real-World Embedded Systems

Embedded systems are defined by how well they sense, process, and respond to the real world. This session explores the complete flow from sensor input to control action, focusing on what makes systems accurate, responsive, and reliable in practice. Through practical examples, we'll examine challenges in sensing, real-time behavior, and control and how these are addressed in modern embedded designs as systems scale in complexity.

Panasonic

Panasonic: The 3 P's: Industry leading products for a variety of markets & applications.

PhotoMOS, Polymers, PIR sensors... there's something unique to say about them all. Whether your need is microseconds switching speed in a miniature (4.6mm³) device, ultra-low ESR for reliable, efficient, filtering in high temp environments (+135°C), or a fully integrated motion sensor with the option of more than 15 different proprietary lenses to choose from. It's all made possible by Panasonic.

TE Connectivity

TE Connectivity Sensing technologies for Industrial, Medical & IoT applications.

Theory of design and application information for various temperature, pressure, position, vibration & force sensing technologies used in industrial, medical and IoT / Condition Monitoring applications.

Session 3

Microchip Technology, Inc.

Simplifying Industrial Automation Design with Multiprotocol Industrial Ethernet and Edge Intelligence

Industrial automation products are expected to support multiple Industrial Ethernet standards, integrate seamlessly with existing control systems, and reach production faster than ever before. In this technical seminar, you'll learn how to simplify the development of industrial controllers, gateways, remote I/O, fieldbus bridges, and HMI-based systems using Microchip's MPU portfolio and RT-Labs' industrial communications software.

We'll explore practical approaches for implementing PROFINET, EtherNet/IP, Modbus TCP, CC-Link, and emerging EtherCAT solutions on a common hardware platform using a unified software architecture. You'll see how RT-Labs' U-Phy multiprotocol framework enables a single application to support multiple industrial networks through a common API, reducing software complexity, development effort, and long-term maintenance.

The session will demonstrate how pre-certified and compliance-tested protocol libraries, pre-compiled software packages, reference applications, and ready-to-use development tools can significantly reduce implementation effort and accelerate certification activities. We'll also examine Microchip's pass-through licensing model, which simplifies procurement by eliminating traditional per-device royalty tracking and software audits while providing a cost-effective path from evaluation to production.

You'll also learn how Linux and RTOS deployment options, integrated security features, and System-in-Package (SiP) and System-on-Module (SOM) solutions can help reduce PCB complexity, lower development risk, and shorten time-to-market. Through live demonstrations on the SAM9 platform, you'll see how to move quickly from evaluation to a working industrial communication solution while maintaining the flexibility to support multiple protocols and evolving Industry 4.0 requirements. Whether you're developing PLCs, gateways, industrial HMIs, remote I/O, or connected edge devices, you'll leave with practical techniques that can help accelerate your next design.

NXP

The Edge Isn't the Wild West. Lock It Down with NXP

As edge devices become more connected and more critical, security can no longer be an afterthought. This session explores how NXP enables you to design secure connected systems from the ground up, while equipping your team with the knowledge and tools to confidently navigate modern security challenges.

TAIYO YUDEN

Taiyo Yuden: Conductive Polymer Hybrid Aluminum Electrolytic Capacitors (Polymer Hybrid Capacitor)

TAIYO YUDEN conductive polymer hybrid aluminum electrolytic capacitors combine the benefits of liquid electrolytes and solid conductive polymers. They offer ultra-low Equivalent Series Resistance (ESR), high ripple current capabilities, and a long operational life at high temp up to 150 °C 2000 hours.

Taiyo Yuden will introduce Polymer Hybrid Capacitor and highlight what the trends are in making quality Hybrid Capacitor. We will contrast our different product families so that you can select the best performance device to support your application and we will cover our website tool for component selection called COMPAS.

Vantron Technology, Inc.

From Edge to Cloud: Edge AI & Networking Solution for Remote Industrial Sites

Explore Vantron's edge-to-cloud solution combining industrial control, embedded computing, IoT networking and cloud management. Through a remote energy deployment, this session showcases how AI-enabled Wi-Fi HaLow IP cameras, HaLow routers, and NXP i.MX94 gateways work together to deliver smart edge processing, kilometer-scale connectivity, secure IIoT monitoring, and seamless cloud integration.

Session 4

MultiTech Systems Inc

Introducing Next-Gen SocketModem®: The Future of Embedded Cellular Connectivity

Join us for a product innovation seminar introducing MultiTech's next-generation, embedded cellular connectivity platform engineered to meet long-term connectivity demands and support seamless technology evolution. Designed to address the critical challenges of cellular-enabled IoT solutions, it simplifies integration, enhances scalability, and accelerates time to market.

In this session, we'll explore the market positioning and showcase its standout features, including Fail-Safe FOTA and Auto-Connect Technology. Whether you're building solutions for smart infrastructure, industrial IoT, or transportation and logistics, the next-generation SocketModem® delivers the scalable, secure, and reliable connectivity your applications demand.

SiliconExpert

Unified Intelligence Turns Supply Chain Volatility into Competitive Advantage

Geopolitical instability, AI-driven demand surges, lifecycle changes, supplier concentration, and market volatility have permanently raised the stakes for electronics manufacturers. The companies that win in this environment act faster and decide smarter, and that requires supply chain, compliance, and engineering to operate from a single source of truth.

Most organizations are not there yet. Fragmented data, low AI confidence, reactive sourcing, and late-arriving signals create bottlenecks that slow every team down and widen the window of exposure. This is an alignment problem as much as a technology problem.

When AI serves as the operating layer of the supply chain, with unified intelligence spanning supply chain, procurement, engineering, and compliance, plus continuous global-event monitoring and mitigation tracking, every team responds to the same prioritized signal at the same time. Immediate BOM-level visibility, AI-summarized risk context, validated alternates, and coordinated procurement response⁴ become the new standard.

This seminar walks through what that integrated model looks like in practice, including live scenario walkthroughs and a clear framework for building resilience that keeps pace with global dynamics.

STMicroelectronics #2

High Voltage Buck and Iso Buck Architectures: Efficient DC/DC Conversion up to 100 V

Join us to dive into modern step down topologies for rugged 12/24/48 V and up to 100 V bus systems using ST's DCPx DC DC buck converters. We will review advanced control techniques (including V2COT), light load optimization strategies (low consumption vs low noise modes), and high frequency operation to reduce magnetics size while maintaining stability and EMC performance. You'll see how iso buck architectures enable compact, transformer isolated supplies without optocouplers, and how to leverage ST's eDesignSuite and eDSim flow to rapidly evaluate efficiency, loop response, and robustness in demanding industrial and automotive designs.

Vishay

Vishay- Current Sense and components for Industrial Motor Control and Power Systems

Vishay will provide an overview of components for industrial motor control and power systems, including: encoders, precision potentiometers, panel potentiometers, medium- and high-power and current sense resistors. The session will focus on product capabilities, key performance characteristics, available technologies, and design considerations, enabling engineers to better understand how these components can address demanding industrial control, feedback, and power management requirements

Session 5

Altera Corporation

Agilex 3/5 FPGAs Low Precision Math Support - Useful for AI and More!

Agenda:

- Overview of the inclusion of Tensor math blocks and Complex Multiplies Natively in Agilex 3/5 DSP blocks
- Tensor FIR / FFT Optimizations to Help Reduce Device Resources
- Using MATLAB to both Design Efficient RTL and Test in Hardware
- Demo

Infineon

Untapping GaN and SiC for the Motor Control World

Learn how Infineon is leveraging their expertise in wide bandgap semiconductors, combined with their newly released PSOC Control microcontrollers, to advance motor control. We will go into the

newest product and packaging solutions along with how to solve the challenges of designing for these new technologies using digital control.

Nuvoton Technology Corporation America

Smart Buzzer® – Transforming Piezo Buzzers into Meaningful Audio Alerts

Stop settling for beeps that users ignore—discover how Nuvoton’s Smart Buzzer® turns conventional piezo buzzers into intelligent voice-capable notification systems that improve safety, usability, and product differentiation while preserving the low cost, power efficiency, and simplicity industrial designers' demand

Session 6

Advanced Energy

Powering FPGAs: Design Considerations and Challenges

This presentation outlines the key challenges of powering FPGA systems, where low voltages and very high currents demand careful design of power architecture, layout, and thermal performance. It highlights the role of DC-DC modules and Point-of-Load (POL) converters in improving efficiency and reducing losses, while addressing critical considerations such as transient response, EMI, current sharing, and inrush control. The session also emphasizes practical trade-offs between modular and discrete designs, along with manufacturing and integration factors, reinforcing that successful FPGA power design requires a balanced, system-level approach.

Critical Link LLC

Critical Link SOMs: In-Depth Implementation of AI at the Edge, Embedded Linux Security, and AMP

Building on last year's introduction to AI at the edge and AMP, this seminar takes a deeper look at real-world implementation. We will cover practical strategies for getting started with edge AI using Altera, Qualcomm, and TI-based SOMs, including a live Agilex 5 demo, techniques for interprocessor communication and resource partitioning across multi-processor and FPGA systems, and an attack surface analysis of embedded Linux systems.

Infineon

Get your motor spinning with PSOC™ Control

In this class we will learn about the new PSOC™ Control family of microcontrollers, the next generation of MCU's developed specifically for motor control and power conversion applications using Si, SiC & GaN. We will do a live demo with the PSOC™ C3M5 series MCU to run an on-the-go motor application using USB pluggable hardware (KIT_PSC3M5_2GO) and ModusToolbox™ Motor Suite software. Customers who bring their laptop and have ModusToolbox loaded on their system are welcome to participate as a hands on demo.

PUI Audio, Inc.

Designing with Audio & Haptics: Practical Techniques to Enhance User Experience

Modern product designs are defined by how users experience them - not just how they function. In this interactive session, PUI Audio demonstrates how audio and haptics improve usability, reinforce brand identity, and enhance safety. Learn practical techniques for selecting, driving, mounting, and validating components through real world examples and demos.